

EE16 Ferrite Core Design Calculations

Core Parameters

$MPL = 3.45 \text{ cm}$	(Magnetic Path Length)
$We = 3.29 \text{ g}$	(Core Weight)
$MLT = 3.40 \text{ cm}$	(Mean Length per Turn)
$A_c = 0.19 \text{ cm}^2$	(Cross-Sectional Area)
$W_a = 0.19 \text{ cm}^2$	(Window Area)
$A_p = A_c \cdot W_a = 0.0361 \text{ cm}^4$	(Core Product)
$K_g = 2.02 \times 10^{-3} \text{ cm}^5$	(Core Geometry)

Design Calculations

1. Current Density (J)

$$J = \frac{2E \cdot 10^4}{B_m \cdot A_p \cdot K_u} = \frac{2 \cdot 4.87 \times 10^{-4} \cdot 10^4}{0.25 \cdot 0.0361 \cdot 0.4} = \frac{9.74}{0.00361} \approx 2698 \text{ A cm}^{-2}$$

2. RMS Current (I_{RMS})

$$I_{\text{RMS}} = \sqrt{I_o^2 + (\Delta I)^2} = \sqrt{4.583^2 + 1.375^2} = \sqrt{21.00 + 1.89} \approx 4.785 \text{ A}$$

3. Bare Wire Area (A_w)

$$A_w = \frac{I_{\text{RMS}}}{J} = \frac{4.785}{2698} \approx 0.00177 \text{ cm}^2$$

- Selected Bare Wire Area:

$$A_{ws} = 0.003 \text{ cm}^2$$

4. Effective Window Area (W_{aeff})

$$W_{\text{aeff}} = W_a \cdot S_3 = 0.19 \cdot 0.75 = 0.1425 \text{ cm}^2$$

5. Number of Turns (N)

$$N = \frac{W_{\text{aeff}} \cdot S_2}{A_{ws}} = \frac{0.1425 \cdot 0.6}{0.003} = \frac{0.0855}{0.003} = 28.5 \approx \boxed{29 \text{ turns}}$$

6. Required Air Gap (L_g)

$$L_g = \frac{0.4\pi N^2 A_c \cdot 10^{-8}}{L} - \frac{MPL}{\mu_0}$$

Substituting values:

$$\begin{aligned} L_g &= \frac{0.4\pi \cdot 29^2 \cdot 0.19 \cdot 10^{-8}}{34.9 \times 10^{-6}} - \frac{3.45}{2500} \\ &= \frac{0.4\pi \cdot 841 \cdot 0.19 \cdot 10^{-8}}{34.9 \times 10^{-6}} - 0.00138 \\ &= \frac{0.4 \cdot 3.1416 \cdot 160.0 \cdot 10^{-8}}{34.9 \times 10^{-6}} - 0.00138 \\ &= \frac{2.01 \times 10^{-6}}{34.9 \times 10^{-6}} - 0.00138 \\ &= 0.0576 - 0.00138 \approx 0.0562 \text{ cm} \end{aligned}$$

Design Notes

- Use **AWG 30** wire ($A_w \approx 0.00177 \text{ cm}^2$)
- Verify core saturation: $B_{\text{sat}} > 0.25 \text{ T}$
- Gap tolerance: Maintain $\pm 0.001 \text{ cm}$ precision
- Thermal check: Ensure $T_r \leq 25^\circ \text{C}$ rise